RF System and Amplifier Design using Keysight ADS

System-Level RF Characterization and Low-Noise Amplifier Design Portfolio

Topics: ACPR/ACLR · EVM · Receiver Link Budget · Spurious Response · IP3/TOI · Transistor Biasing · S-Parameters · Noise Figure · Stability · Smith Chart Matching · Input/Output L-C Matching

Tools: Keysight Advanced Design System (ADS)

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Abstract

This report presents a two-part radio-frequency (RF) design study carried out in Keysight Advanced Design System (ADS). The first part characterizes an RF amplifier and receiver chain at the system level, quantifying linearity and signal quality through adjacent-channel power ratio (ACPR) and error-vector magnitude (EVM) under a π/4-DQPSK modulated source, followed by a cascaded receiver link-budget, spurious-response mapping, and two-tone third-order intercept (IP3) analysis with and without intermediate-frequency (IF) filtering. The second part addresses the circuit level: a single-transistor amplifier is biased, its device model is verified against measured S-parameters, its noise figure and stability are analyzed using source-pull noise circles, and input/output impedance-matching networks are synthesized. The receiver chain achieves ≈ 76.1 dB cascaded gain at ≈ 4.2 dB noise figure- reproduced by a worked Friis cascade that lands at 3.7 dB- with a measured two-tone OIP3 ≈ 21.8 dBm (IIP3 ≈ −25.7 dBm); the amplifier reaches a minimum noise figure NF_{min} ≈ 1.74 dB near 1.86 GHz, and is stabilized and matched with a synthesized input/output L-C network. Each result is interpreted against closed-form RF relations and cross-checked by hand where the available data supports it.

Index Terms- ACPR, EVM, link-budget, Friis formula, low-noise amplifier (LNA), noise figure, third-order intercept (IP3), S-parameters, Rollett stability, noise circles, impedance matching, Keysight ADS.

Design area	Method / metric	Outcome
Transmitter linearity	ACPR / ACLR with pi/4-DQPSK	Adjacent-channel spectral regrowth evaluated; more negative ACPR indicates better leakage performance
Signal quality	EVM and constellation error	Ideal and distorted symbol trajectories compared; RMS EVM equation defined.
Receiver chain	Cascaded gain and noise figure	Approx. 76.1 dB gain and approx. 4.2 dB cascaded NF from visible ADS labels.
Interference	Spurious + two-tone IP3	Spurious products and IM3 terms identified; IF filtering suppresses post-generation out-of-band products.
Device design	Spurious products and IM3 terms identified; IF filtering suppresses post-generation out-of-band products.	Bias point selected and model checked before noise/matching design.
LNA design	Noise circles + stability + source match	NFmin approx. 1.744 near 1.86 GHz; Sopt and source-match tradeoff extracted.
Power transfer	Output L/C matching	Output network synthesized
Matching network	Input/output L-C matching and stabilization	Cin 30 pF, Lin 4 nH, Rstab 39 ohm, Lstab 25 nH, Lout 16 nH, Cout 2 pF
Power transfer	Input/output L-C matching	Stabilization and matching elements summarized from ADS screenshots.

Table 1. Summary of the design tasks, methods, and measured outcomes.

1. Introduction and Design Flow

Modern RF front-ends face two competing requirements. They must be linear enough that a digitally-modulated signal is neither distorted within its own band nor allowed to regrow into adjacent channels, and sensitive enough to recover weak desired signals against the noise floor while tolerating strong interferers. The first is governed by large-signal metrics- ACPR, EVM, and the third-order intercept; the second by the noise figure and the cascaded dynamic-range budget. This study uses Keysight ADS to address both, progressing from behavioural system models (Sections 2–6) to a transistor-level amplifier (Sections 7–11): biasing, model verification, noise and stability, and impedance matching.

Design activity	**Purpose**
ACPR analysis	Quantify adjacent-channel spectral regrowth from non-linearity
EVM analysis	Measure modulated-signal quality against the ideal constellation
Receiver link-budget	Cascade gain, noise figure, and intercept to predict sensitivity
Spurious-response analysis	Identify in-band mixing products that mimic signal
Third-order intercept (IP3)	Quantify two-tone intermodulation and dynamic range
Transistor bias setup	Fix the DC operating point of the amplifier
S-parameter model verification	Validate the device model against datasheet data
Noise & stability analysis	Determine NFmin, stability, and the optimum source match
Impedance matching	Synthesize input and output matching networks

Table 2. Design activities and their purpose in the RF design flow.

2. Adjacent-Channel Power Ratio (ACPR)

ACPR (equivalently ACLR) is the dominant linearity metric for digitally-modulated transmitters. It is the ratio of the integrated power in an adjacent channel to that in the main channel, reported in dBc:

$$\mathrm{ACPR} = 10 \log_{10} \left(P_{adj} / P_{main} \right), \quad P_{ch} = \int_{ch} S(f)\, df \tag{1}$$

where S(f) is the power spectral density and the integral runs over each channel bandwidth. A more negative ACPR denotes better leakage performance. Spectral regrowth into the adjacent channel is a direct consequence of amplifier non-linearity acting on a non-constant-envelope signal: expanding the amplifier transfer characteristic as a power series $y = a_1x + a_2x^2 + a_3x^3 + \ldots$, the third-order term a_3 both compresses the gain and generates the out-of-band regrowth that ACPR measures. The test bench of Figure 1 drives the amplifier with a π/4-DQPSK source and probes the upper, main, and lower channel powers; Figure 2 adds receive-side filtering so ACPR can be evaluated including channel-select effects.

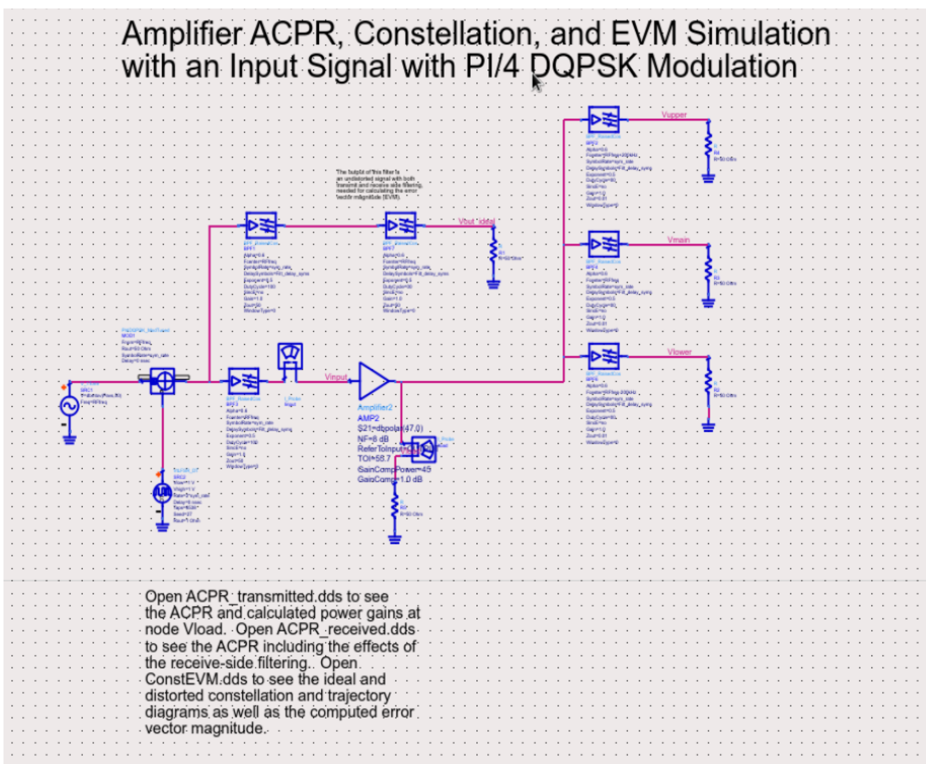

Figure 1. ACPR / constellation / EVM test bench- RF amplifier driven by a π/4-DQPSK source with upper, main, and lower channel-power probes.

The schematic shows the envelope-simulation test bench. The modulated source and channel-power meters let ADS integrate P_{main} and P_{adj} directly per Eq. (1); placing meters on both the transmit and filtered receive nodes separates raw amplifier regrowth from the channel-selectivity that filtering later recovers.

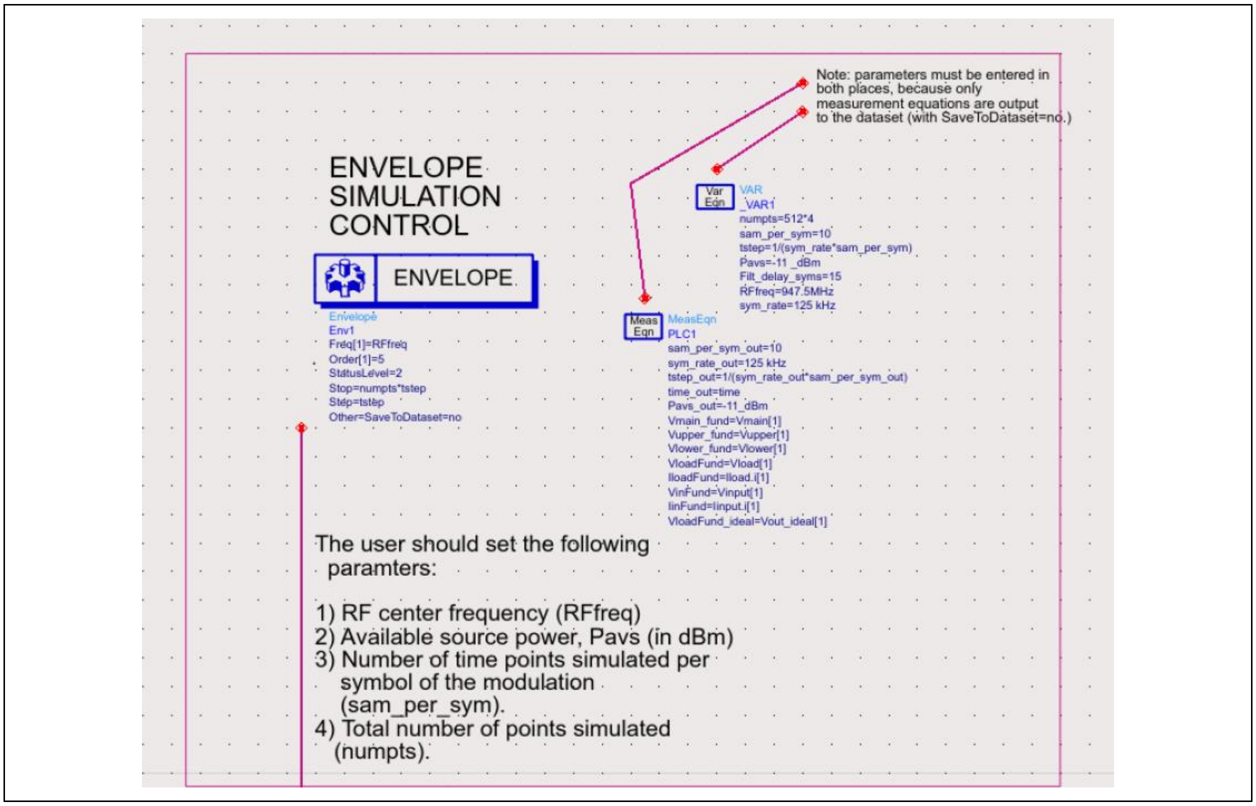

Figure 2. Envelope-simulation control and ACPR continuation block.

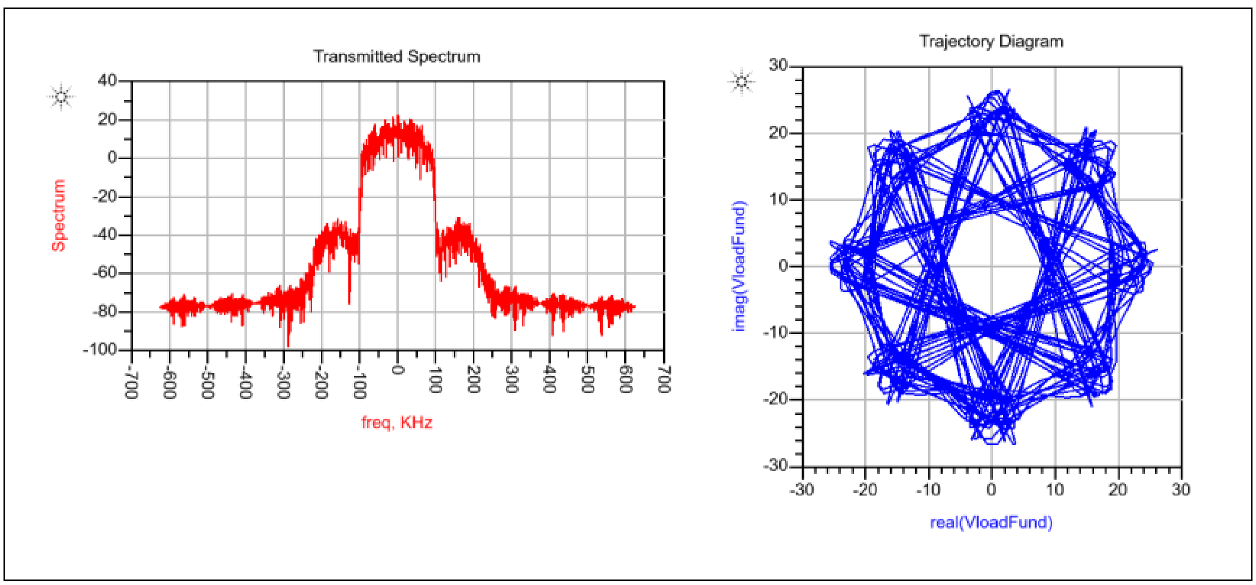

Figure 3. Transmitted spectrum and trajectory diagram- adjacent-channel regrowth and the I/Q signal trajectory.

The plot shows the transmitted spectrum and the I/Q trajectory. The regrowth lobes flanking the main channel sit ≈ 50–60 dB below the main-lobe peak, the spectral signature of third- and higher-order distortion; the trajectory diagram traces the modulated signal through the I/Q plane, where deviation from the ideal path is the time-domain face of the same non-linearity.

3. Error-Vector Magnitude (EVM)

Where ACPR looks outward at neighbouring channels, EVM looks inward at the wanted signal. For each received symbol the error vector is the difference between the measured and ideal constellation points; the RMS EVM normalizes the error power to the reference power over all N symbols:

$$EVM\_RMS(\%) = \sqrt{ \sum_k |S_{meas,k} - S_{ideal,k}|^2 / \sum_k |S_{ideal,k}|^2 } \times 100\% \quad (2)$$

where $S_{meas,k}$ is the received or distorted symbol and $S_{ideal,k}$ is the ideal reference symbol. EVM captures the combined effect of non-linearity, noise, phase error, amplitude imbalance, and filtering, making it one of the most informative quality metrics for a modulated link. For a noise-limited link, $EVM\_RMS \approx 1/\sqrt{SNR}$, so an EVM floor maps directly onto a sensitivity floor.

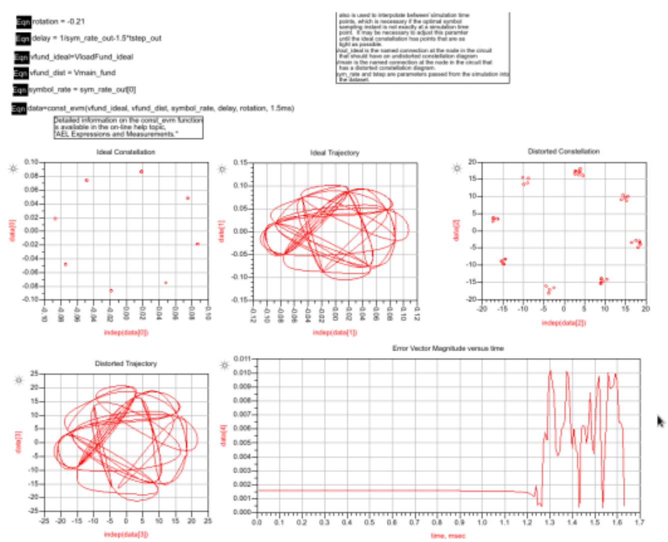

Figure 4. EVM evaluation- distorted versus ideal constellation.

The plot compares the distorted and ideal constellations. The peak error-vector magnitude is ≈ 0.011 in normalized units; the tightness of the symbol clusters around their ideal points is the visual measure of EVM, and any spreading marks the onset of distortion that sets the practical input-power ceiling.

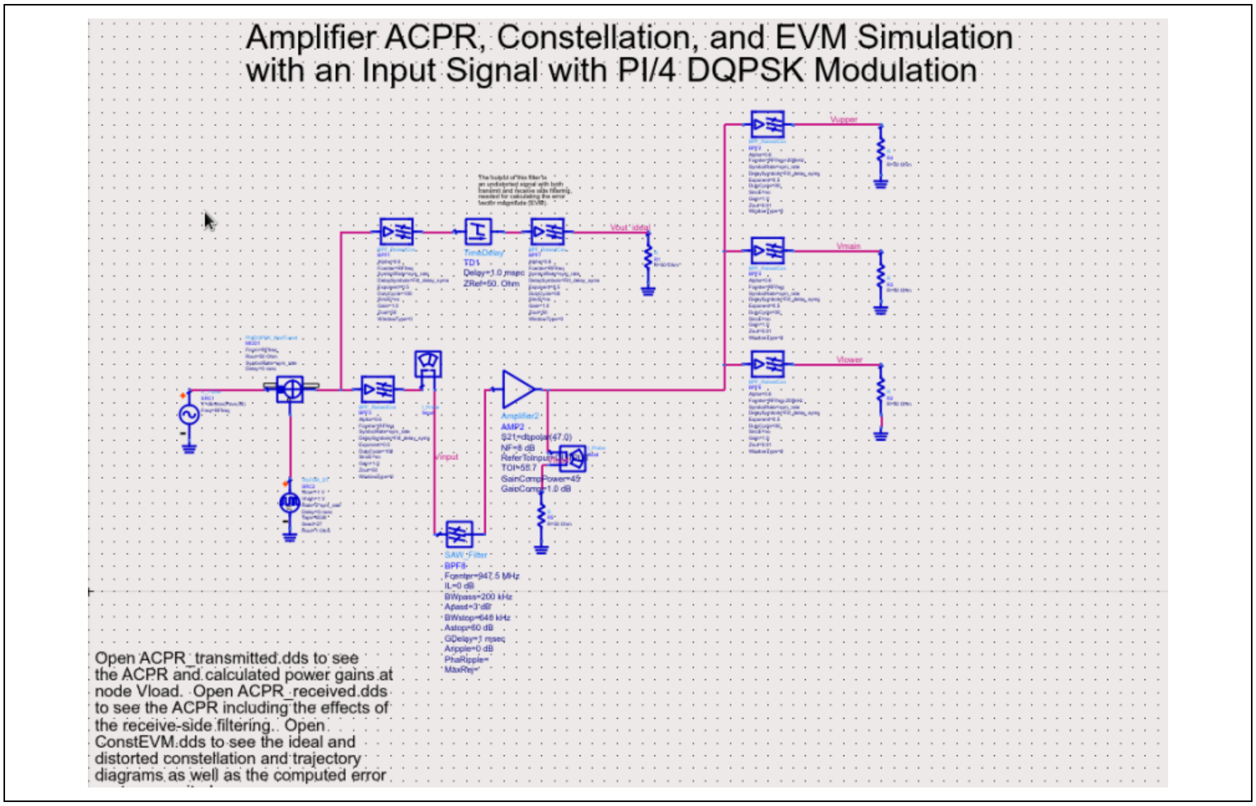

Figure 5. EVM schematic- ideal and distorted paths differenced to compute error-vector magnitude.

The schematic shows the measurement method: an ideal reference path and the amplifier path are differenced symbol-by-symbol to form e_k, which Eq. (2) reduces to a single RMS figure.

4. Receiver Link-Budget Analysis

A receiver budget cascades the gain, noise figure, and intercept of every stage to predict end-to-end sensitivity and dynamic range before any circuit is built. The cascaded noise factor follows Friis's formula, evaluated in linear terms (noise factors F, power gains G), not dB:

$$F_{tot} = F_1 + (F_2-1)/G_1 + (F_3-1)/(G_1G_2) + \ldots \tag{3}$$

Because each successive term is divided by the cumulative gain ahead of it, the first stage dominates F_{tot} — the rationale for a low-noise, high-gain front end. The total gain is the product (sum in dB) of the stage gains, and the sensitivity follows from the noise floor:

$$P_{sens} = -174\ \text{dBm/Hz} + 10\log_{10}B + NF + SNR_{min} \tag{4}$$

where B is the noise bandwidth and −174 dBm/Hz is the thermal-noise density kT at 290 K. From the ADS budget controller the cascaded gain is ≈ 76.1 dB and the cascaded noise figure ≈ 4.2 dB. Figure 6 shows the per-stage specifications feeding the cascade.

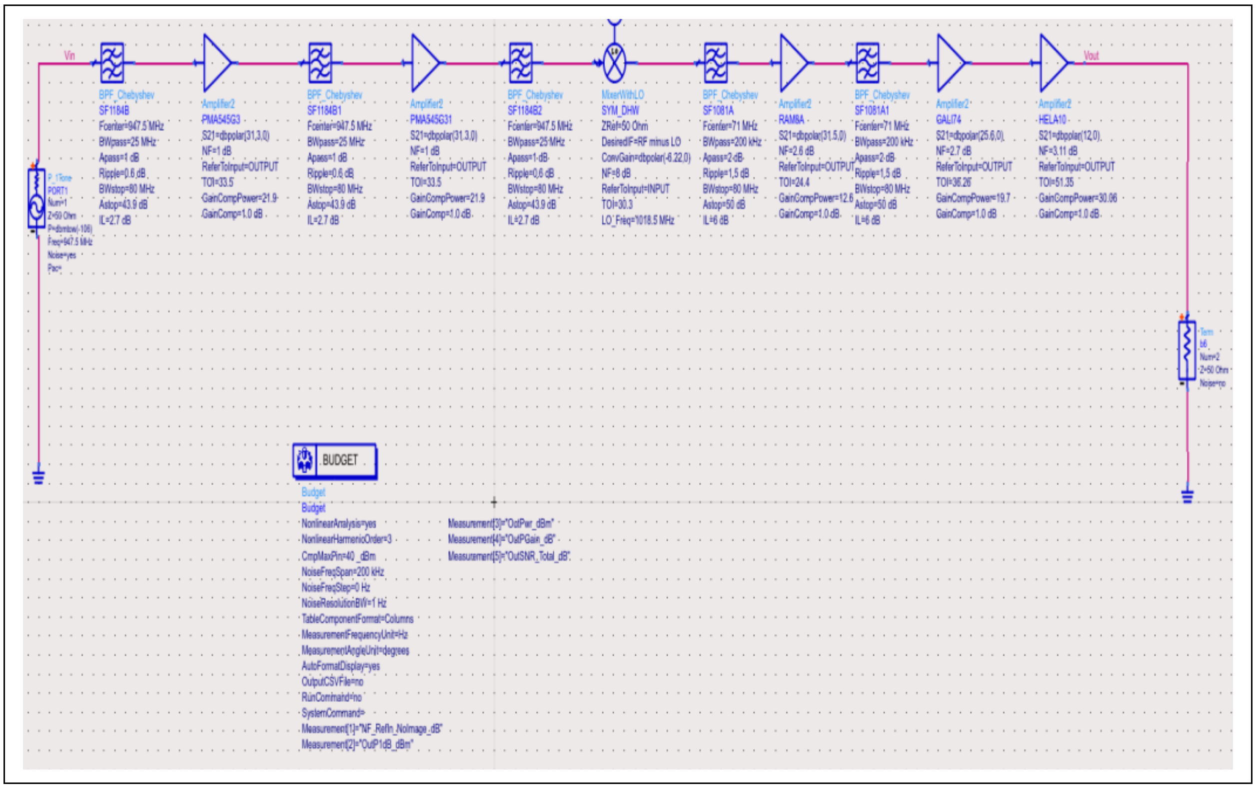

Figure 6. Receiver specifications for budget analysis- per-stage gain, NF, and intercept feeding the ADS budget controller.

The diagram shows the cascaded receiver: RF filtering, gain blocks, a mixer, IF filtering, and IF/baseband gain. The first RF filter and amplifier dominate the cascaded NF, consistent with Friis (Eq. 3); the visible stage labels give the gain, NF, and intercept that sum to the ≈ 76.1 dB / 4.2 dB system result. Budget analysis is where feasibility is decided- NF allocated to the front, linearity to the back, and the cascade verified to close against sensitivity (Eq. 4).

4.1 Worked Friis cascade from the stage parameters

To confirm that the system noise figure follows from the device physics rather than the budget tool alone, the cascade is reproduced by hand from the per-stage parameters annotated in Figure 6. The chain comprises eleven stages- alternating band-pass filters (modelled by their insertion loss IL, with NF = IL) and gain blocks- with a down-converting mixer in the middle. Table 3 lists the stage gains and noise figures read from the schematic, together with each stage's contribution to the total noise factor, $(F_i-1)/G_{cum}$, where G_{cum} is the cumulative linear gain preceding that stage.

Stage	Gain (dB)	NF (dB)	ΔF contribution	Running NF (dB)
BPF 1	−2.7	2.7	1.862	2.70
Amp 1 (PMA545G3 mode1)	+31.3	1.0	0.482	3.70
BPF 2	−2.7	2.7	0.001	3.70
Amp 2 (PMA545G3 mode1)	+31.3	1.0	0.001	3.70
BPF 3	−2.7	2.7	≈0	3.70
Mixer (SYM-DHW)	−6.2	8.0	≈0	3.70
BPF 4	−6.0	6.0	≈0	3.70
Amp 3 (RAM8A mode1)	+31.5	2.6	≈0	3.70
BPF 5	−6.0	6.0	≈0	3.70
Amp 4 (GALI74 mode1)	+25.6	2.7	≈0	3.70
Amp 5 (HELA10 mode1)	+12.0	3.1	≈0	3.70

Table 3. Worked Friis cascade- stage parameters from Figure 6 and their cumulative noise-figure contribution.

Applying Eq. (3) in linear terms gives a cascaded noise factor $F_{tot} \approx 2.35$, i.e. a noise figure of ≈ 3.7 dB, in good agreement with the ≈ 4.2 dB reported by the ADS budget controller; the residual ≈ 0.5 dB is accounted for by the input-filter insertion loss ahead of the first active stage and by reading tolerance on the schematic labels. The calculation makes the Friis principle explicit: the first band-pass filter and the first amplifier together fix the noise figure at 3.70 dB, and every subsequent stage adds essentially nothing$F_i-1)/G_{cum}$ collapses below 0.001 because the ≈ 31 dB of gain in the first amplifier divides it down. This is precisely why front-end loss is so costly and why the first amplifier must be low-noise and high-gain: a decibel of insertion loss added before it would raise the system NF almost decibel-for-decibel, whereas the entire back end is essentially noise-transparent.

5. Spurious-Response Analysis

Mixers and non-linear stages generate spurious products at frequencies $m \cdot f_{RF} \pm n \cdot f_{LO}$ for integers m, n. If such a product falls within the IF bandwidth it can appear as a false wanted signal. Mapping these responses (Figure 7) identifies the vulnerable frequencies and informs filter placement and frequency planning.

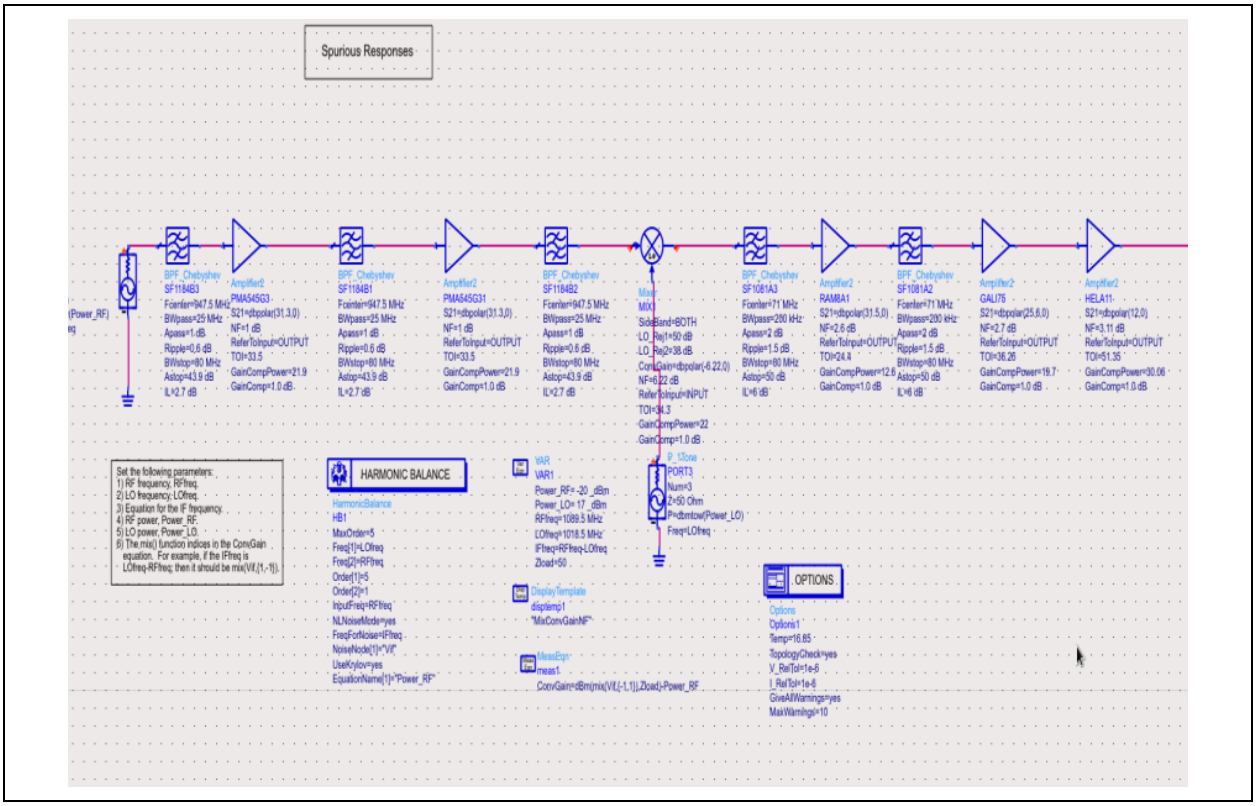

Figure 7. Schematic for spurious-response analysis.

The schematic sweeps the input across frequency and records where mixing products land in-band. The dangerous responses are the low-order (small m, n) spurs, since these are strongest; the analysis tells the designer which must be rejected by RF pre-selection before the mixer.

6. Third-Order Intercept (IP3 / TOI)

When two closely-spaced tones at f_1 and f_2 pass through a non-linear stage, third-order intermodulation generates products at $2f_1-f_2$ and $2f_2-f_1$, immediately adjacent to the originals and therefore impossible to filter. Because the IM3 products rise at three times the rate (in dB) of the fundamental, the two extrapolated lines intersect at the third-order intercept point:

$$\mathrm{IIP3} = P_{in} + \Delta P / 2, \quad \Delta P = P_{fund} - P_{IM3} \tag{5}$$

$$\mathrm{OIP3} = \mathrm{IIP3} + G, \quad \mathrm{SFDR} = (2/3)(\mathrm{IIP3} - N_{floor}) \tag{6}$$

where ΔP is the decibel spacing between fundamental and IM3, G the stage gain, and SFDR the spurious-free dynamic range. A higher IP3 denotes a more linear stage and a wider dynamic range. Figures 8–10 show the two-tone test bench and the IM3 products for tones at 70/72 MHz and 69/73 MHz.

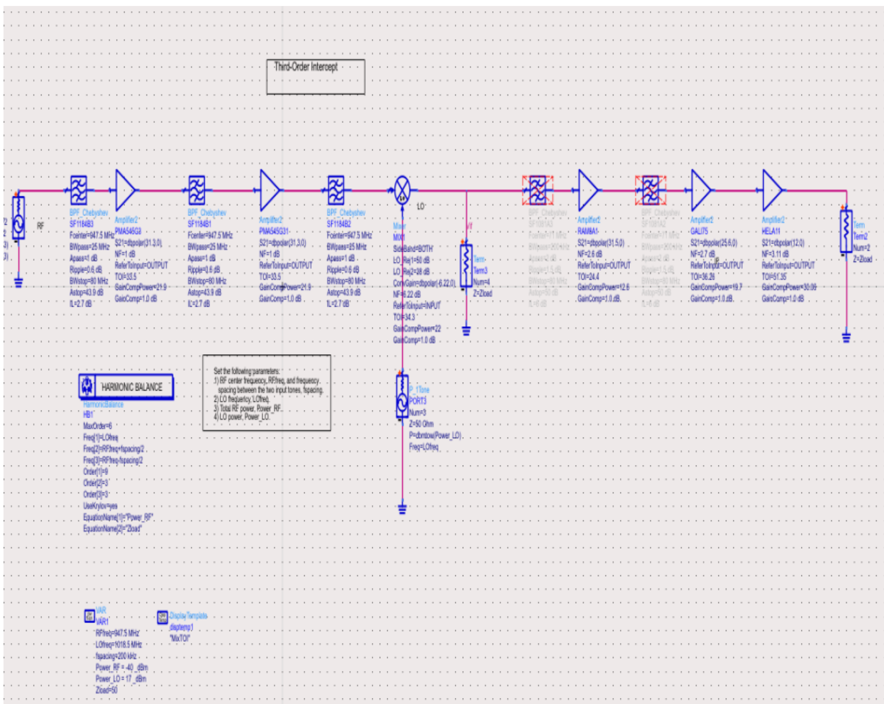

Figure 8. Two-tone third-order-intercept test bench.

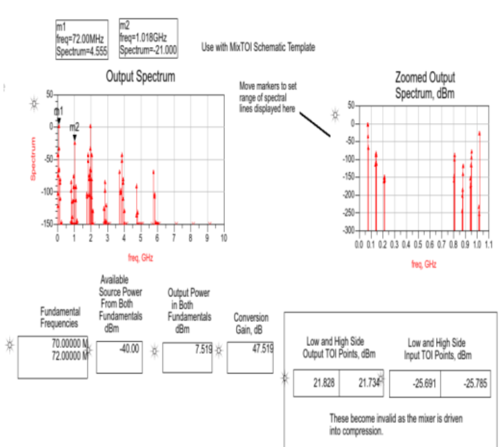

Figure 9. TOI result, tones at 70 and 72 MHz.

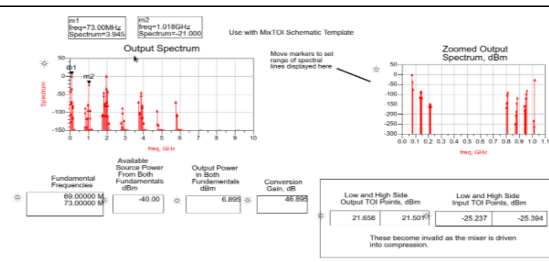

Figure 10. TOI result, tones at 69 and 73 MHz.

The plots show the two fundamentals and the third-order intermodulation products. From the ADS markers, the fundamentals at 70 and 72 MHz are driven at −40 dBm source power and appear at +7.5 dBm output, giving a conversion gain of ≈ 47.5 dB; the extrapolated intercepts are OIP3 ≈ 21.8 dBm and IIP3 ≈ −25.7 dBm. These satisfy Eq. (6) exactly — OIP3 − IIP3 = 47.5 dB equals the conversion gain- confirming the extraction is self-consistent. The negative IIP3 is expected for a high-gain receiver chain referred to its input, where the large cascade gain pushes the input-referred intercept well below 0 dBm; the ADS note that the figures "become invalid as the mixer is driven into compression" correctly bounds the validity of the linear-extrapolation method to the small-signal region. The closer the IM3 products sit to the noise floor, the wider the spur-free dynamic range of Eq. (6).

To demonstrate the benefit of channel filtering, the tone spacing was varied and an IF filter introduced (Figure 11).

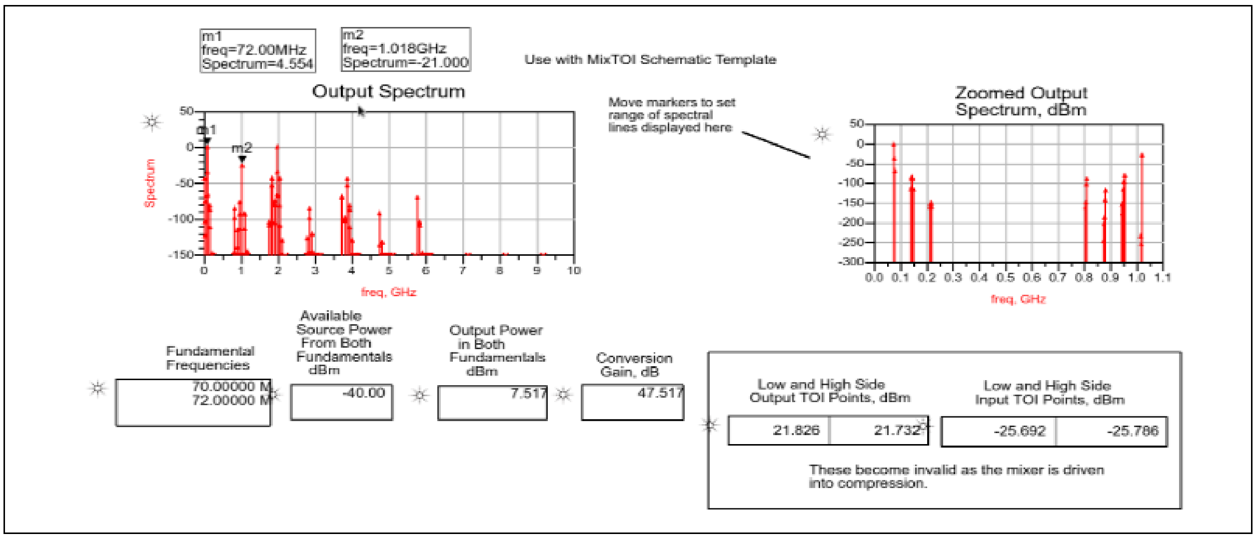

Figure 11. TOI result, tones at 70 and 72 MHz, after IF filtering.

The plot shows the same two-tone test with the IF filter active. Filtering cannot remove in-band IM3 products and does not improve the intrinsic linearity of the nonlinear device. However, it can suppress out-of-band mixing products after they are generated, improving post-filter interference rejection and widening the usable dynamic range seen by the following IF/baseband stages.

7. Transistor Bias Setup

Part II proceeds to a physical transistor. The DC operating point simultaneously sets the small-signal gain, the noise figure, and the linearity, so it must be established before any small-signal design. The bias current fixes the transconductance g_m and hence the available gain and the transition frequency f_T:

$$g_m = \partial I_{out}/\partial V_{in}, \quad f_T \approx g_m / (2\pi C_{in}) \tag{7}$$

written here in device-neutral form: g_m is the small-signal transconductance ($\partial I_{out}/\partial V_{in}$, i.e. $\partial I_C/\partial V_{BE} \approx I_C/V_T$ for the bipolar MMIC gain blocks used in this receiver, or $\partial I_D/\partial V_{GS}$ for a FET), and C_{in} the effective input capacitance that limits the device transition frequency. The relation states the fundamental speed–current trade: raising the bias current increases g_m and f_T, improving gain and bandwidth at the cost of power and, eventually, linearity.

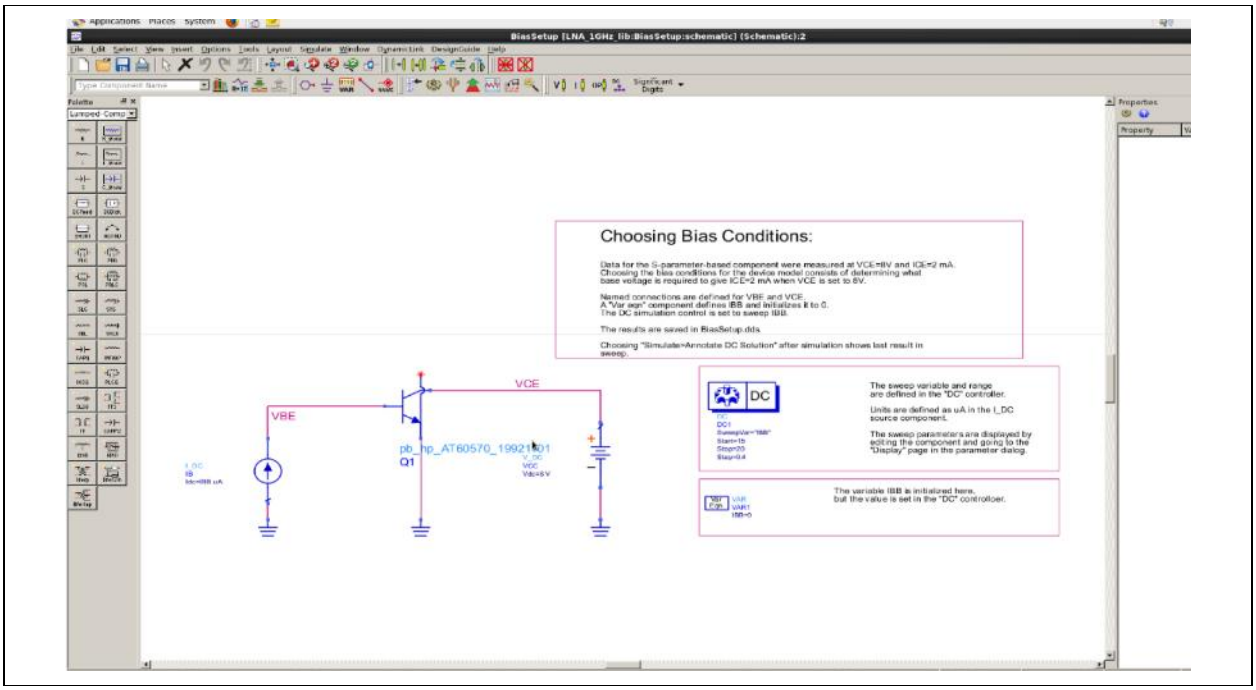

Figure 12. Bias-setup schematic.

The schematic shows the DC bias network that fixes the operating point. Bias determines transconductance, gain, noise figure, and compression behaviour together, so it is selected first; the optimized network reports an operating point near $V_{CE} \approx 8.05$ V and $I_{CE} \approx 2.03$ mA.

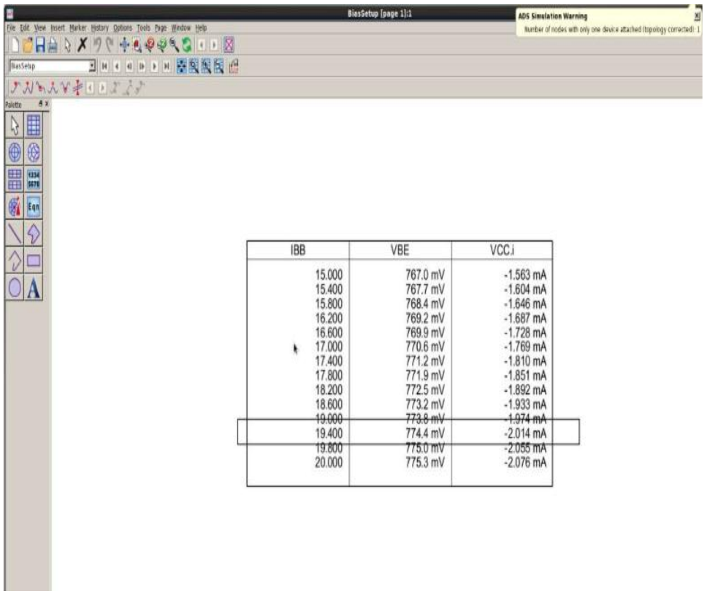

Figure 13. Bias-setup simulation table.

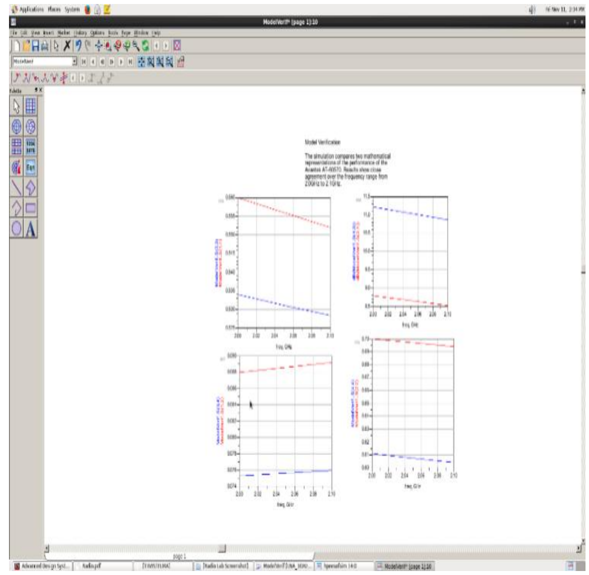

Figure 14. Bias-setup result — DC operating-point verification.

The tables show the solved operating point. Confirming the bias before S-parameter, noise, and matching analysis ensures all subsequent small-signal results are referenced to a stable, repeatable point.

8. Model Verification and S-Parameters

Before a transistor model is trusted, its simulated S-parameters are compared against the manufacturer's measured data. The scattering parameters relate the incident (a) and reflected (b) travelling waves at the ports, b = S a, and are the basis for gain, return-loss, stability, noise, and matching analysis:

$$S_{11} = b_1/a_1\,|_{a_2=0},\quad S_{21} = b_2/a_1\,|_{a_2=0},\quad RL = -20\log_{10}|S_{11}| \tag{8}$$

The transducer power gain combines the device S-parameters with the source and load reflection coefficients:

$$G_T = \left[\, (1-|\Gamma_S|^2)|S_{21}|^2(1-|\Gamma_L|^2) \,\right] / \left|\, (1-S_{11}\Gamma_S)(1-S_{22}\Gamma_L) - S_{12}S_{21}\Gamma_S\Gamma_L \,\right|^2 \tag{9}$$

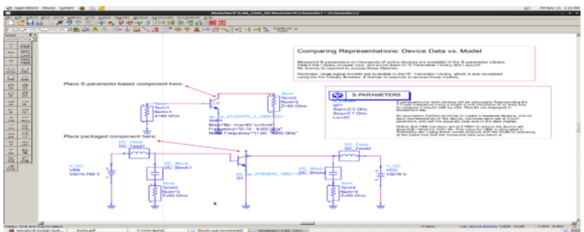

Figure 15. Model-verification schematic.

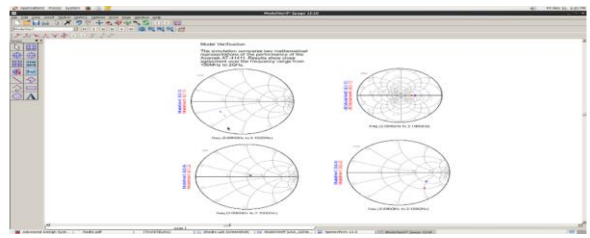

Figure 16. Smith-chart S-parameter comparison.

The schematic and Smith chart show the model checked against reference data. Agreement on the Smith chart confirms the input/output reflection (S_{11}, S_{22}) is reproduced; the trajectory's frequency progression also reveals the device's reactive behaviour, which the matching network will later cancel.

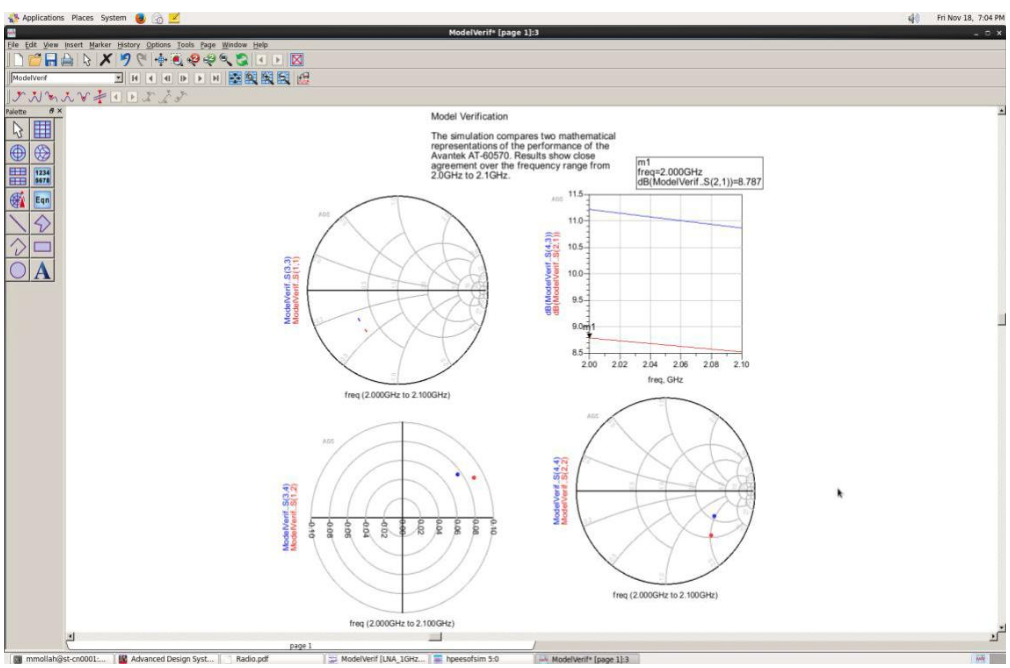

Figure 17. Rectangular S-parameter plots — magnitude and phase of S11, S21, S12, S22 versus frequency.

The plots show the four S-parameters across the band. $|S_{21}|$ gives the forward gain, $|S_{11}|$ and $|S_{22}|$ the input/output match, and $|S_{12}|$ the reverse isolation that governs stability through Eq. (10)

below; the rectangular view exposes frequency-dependent deviations more clearly than the Smith chart.

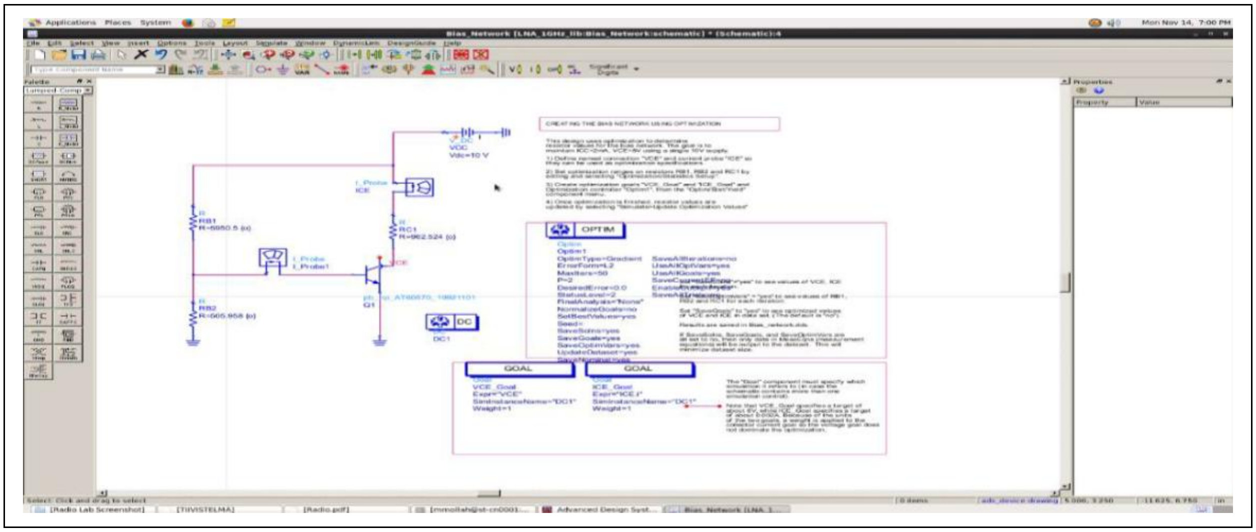

Figure 18. Combined S-parameter result plots — Smith-chart and rectangular views jointly summarizing match, gain, and isolation.

9. Bias-Network Design, Noise and Stability

A practical RF amplifier must isolate the DC bias path from the RF signal path — bias tees, RF chokes, bypass capacitors, and stabilization components let the transistor receive its operating point while the RF ports still see the intended source and load impedances. Figure 19 shows the bias-network schematic and Figure 21 its simulated response. Stability is then assessed with the Rollett factor K and the determinant Δ:

$$K = (1 - |S_{11}|^2 - |S_{22}|^2 + |\Delta|^2) \,/\, (2|S_{12}S_{21}|), \quad \Delta = S_{11}S_{22} - S_{12}S_{21} \tag{10}$$

$$\text{Unconditional stability:} \quad K > 1 \quad \text{and} \quad |\Delta| < 1 \quad (\equiv \mu > 1) \tag{11}$$

where the single-parameter μ factor (μ > 1) is an equivalent, often-preferred test. Stabilization resistors are added where K ≤ 1 to force unconditional stability across the band, at a small cost in noise figure and gain.

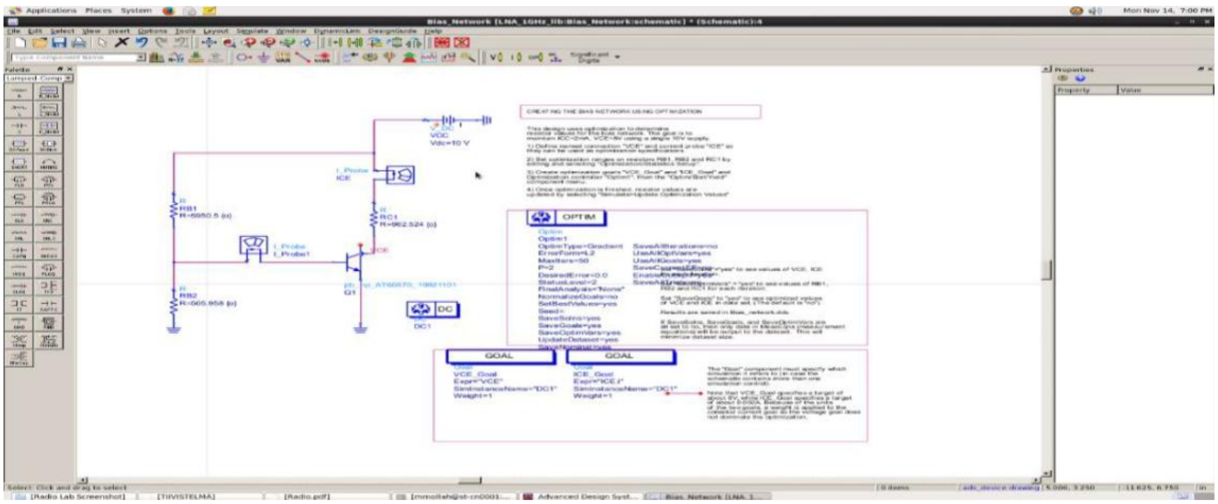

Figure 19. Bias-network schematic- provides the operating point while separating RF and DC paths.

The schematic shows the bias network and stabilization elements. The optimized network reports $V_{CE} \approx 8.05$ V and $I_{CE} \approx 2.03$ mA; the series/shunt stabilization resistors raise K above unity, the explicit trade of a little gain and noise for guaranteed stability per Eq. (11).

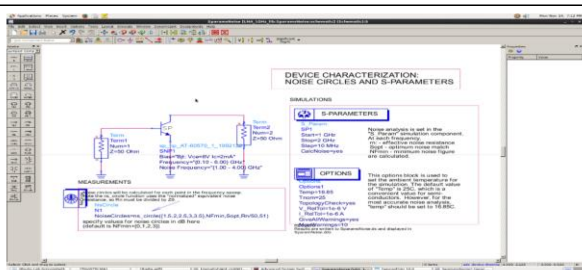

Figure 20. S-parameter / noise-analysis setup.

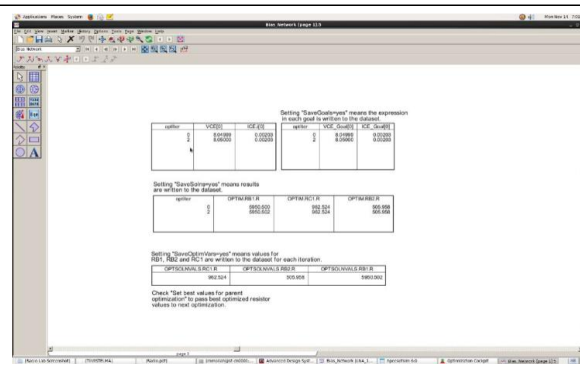

Figure 21. Bias-network simulation result.

The setup prepares the circuit for S-parameter, noise, stability, and matching evaluation, and the result confirms the bias behaviour used downstream. Evaluating K and |Δ| over the whole band not just the design frequency — is essential, since amplifiers are most prone to instability outside the intended band.

10. Noise Figure, Optimum Source Match and Noise Circles

An LNA is defined by the trade-off between minimum noise figure and maximum available gain, which generally occur at different source impedances. The noise figure as a function of the source reflection coefficient Γ_S follows the classical relation

$$F = F_{min} + (4R_n/Z_0) \cdot |\Gamma_S - \Gamma_{opt}|^2 \,/\, [\,(1-|\Gamma_S|^2)\,|1+\Gamma_{opt}|^2\,] \qquad (12)$$

where F_{min} is the minimum noise factor, R_n the equivalent noise resistance, and Γ_{opt} the optimum source reflection. Setting F constant traces **constant-noise-figure circles** on the Γ_S plane; the designer chooses Γ_S to balance noise against gain while keeping $K > 1$. Because $\Gamma_{opt} \neq \Gamma_S^*$ (the conjugate-gain match), the LNA design is an explicit compromise among NF, gain, stability, and return loss.

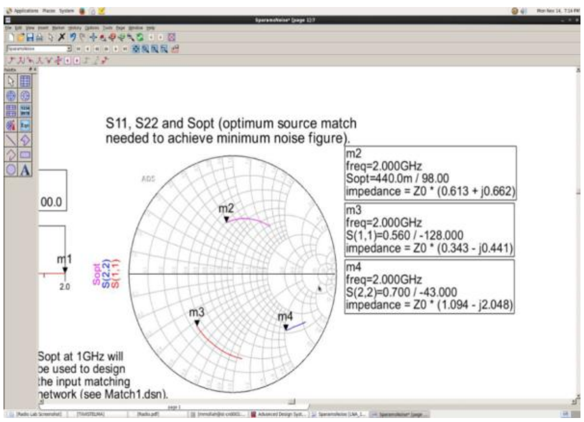

Figure 22a. S11, S22 and optimum source match.

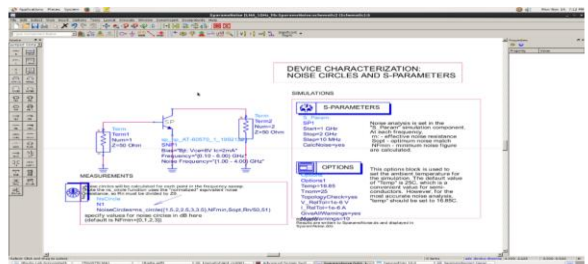

Figure 22b. S-parameter / noise-analysis schematic.

The Smith chart shows S_{11}, S_{22}, and the optimum-noise point Γ_{opt}. The separation between Γ_{opt} and the conjugate-match point is precisely the NF-versus-gain trade-off; the chosen source match sits between them.

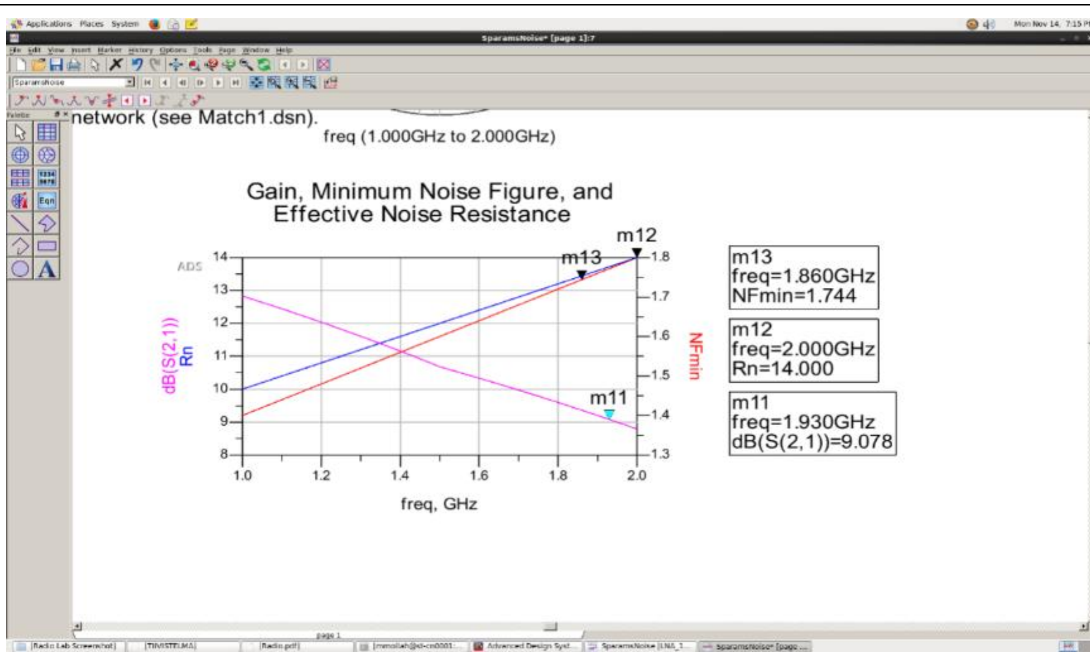

Figure 23. Gain, minimum noise figure, and effective noise resistance versus frequency.

The plot shows NF_{min}, associated gain, and R_n across the band. The minimum noise figure is ≈ 1.74 dB near 1.86 GHz; R_n quantifies how sharply NF degrades as Γ_S departs from Γ_{opt} - a small R_n gives a forgiving design with widely-spaced noise circles.

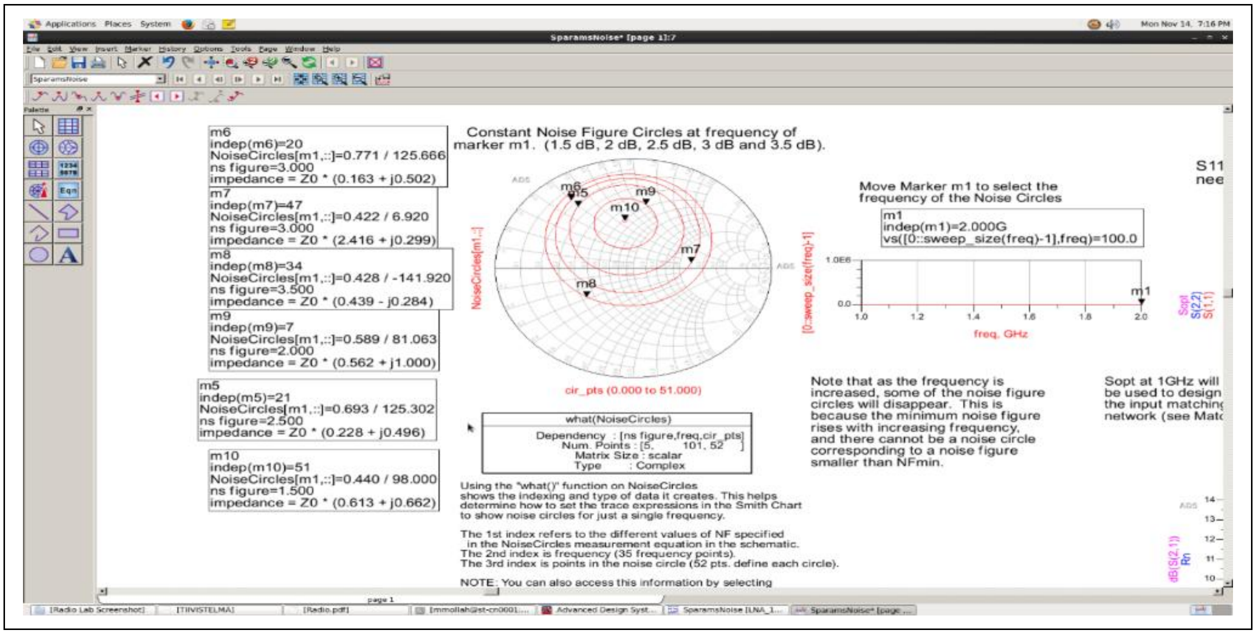

Figure 24. Constant noise-figure circles on the source-impedance plane.

The noise circles geometrize Eq. (12): each circle is a locus of constant NF, centred on Γ_{opt}. Selecting Γ_S on a circle slightly larger than NF_{min} trades a few tenths of a dB of noise for a better gain and input match- the heart of LNA design.

11. Impedance-Matching Network Synthesis

The final step matches the source to Γ_{opt} and the load to the device output for maximum power transfer and flat gain. For a two-element (L-section) match between resistances R_{high} and R_{low}, the loaded quality factor and element reactances are

$$Q = \sqrt{(R_{high}/R_{low} - 1)}, \quad X_{series} = Q\,R_{low}, \quad X_{shunt} = R_{high}/Q \qquad (13)$$

from which the inductor and capacitor values follow at the design frequency via $X_L = \omega L$ and $X_C = 1/\omega C$. The synthesized network uses an input shunt-C / series-L section and an output series-L / shunt-C section, with series stabilization, giving the element set Cin ≈ 30 pF, Lin ≈ 4 nH, Rstab ≈ 39 Ω, Lstab ≈ 25 nH, Lout ≈ 16 nH, Cout ≈ 2 pF.

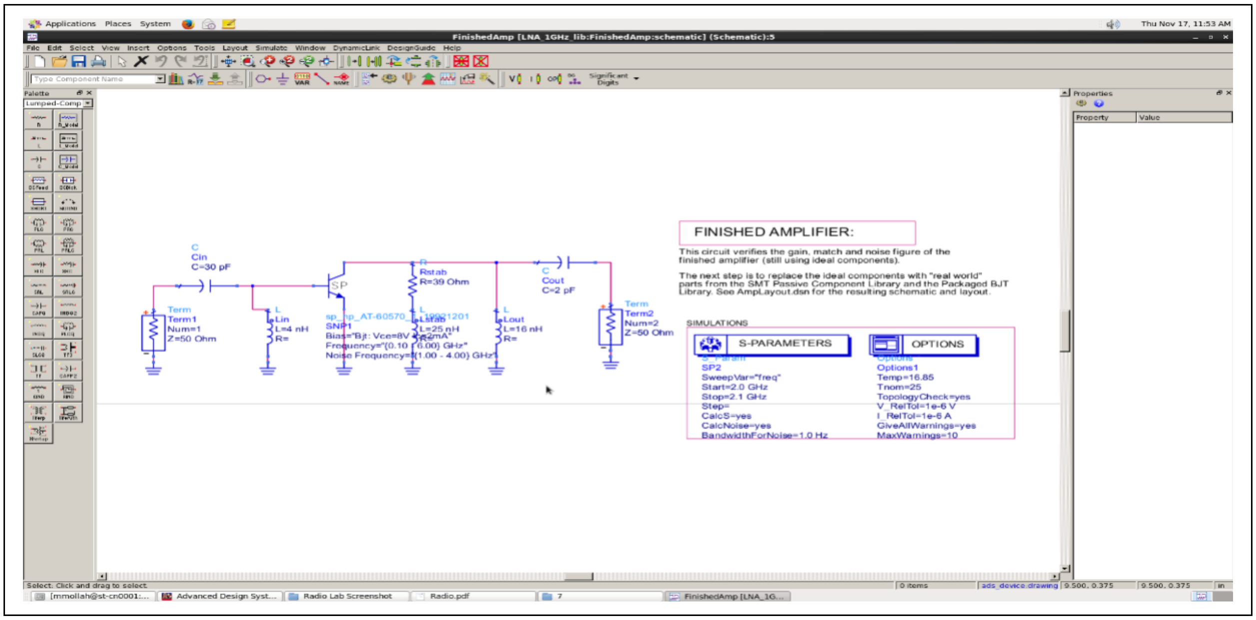

Figure 25. Input/output matching network - schematic and synthesized L-C element values.

The schematic shows the completed matched amplifier. The input network transforms 50 Ω toward Γ_{opt} for minimum noise, while the output network conjugately matches the device output for maximum power transfer; together with the stabilization elements they realize a biased, model-verified, noise-optimized, unconditionally-stable, and impedance-matched LNA.

Element	**Value**	**Function**
Cin	≈ 30 pF	Input shunt - transform toward Γopt
Lin	≈ 4 nH	Input series - noise match
Rstab	≈ 39 Ω	Stabilization - force K > 1
Lstab	≈ 25 nH	Stabilization choke
Lout	≈ 16 nH	Output series- load match
Cout	≈ 2 pF	Output shunt- power match

Table 4. Synthesized matching and stabilization network elements.

12. Measurement Methodology and Bench Correspondence

Each simulated analysis in this report has a direct hardware counterpart. This section maps every ADS result to the bench instrument and technique that measures the same quantity, demonstrating that the design is framed in measurable terms and that the path from simulation to a measured prototype is understood end to end.

12.1 Simulation-to-instrument correspondence

Simulated analysis (this report)	**Bench instrument**	**Measurement technique**
S-parameters / gain / match (Figs. 16–18)	Vector network analyzer (VNA)	S11/S21/S12/S22 after SOLT/TRL calibration; gain, return loss, matching, and stability extraction
ACPR / spectrum (Fig. 3)	Spectrum / signal analyzer	Channel-power integration and ACLR/ACPR marker extraction
EVM / constellation (Fig. 4)	Vector signal analyzer (VSA)	Demodulated EVM, constellation, trajectory, and modulation-quality analysis
Two-tone IP3 (Figs. 9–11)	2× signal generators + spectrum analyzer	Two-tone test; IM3 spacing extraction; IIP3/OIP3 calculation
Noise figure / NFmin (Fig. 23)	Noise-figure analyzer / Y-factor	Y-factor noise-figure measurement; source-pull extraction for extraction for NFmin, Rn, Γopt and Gamma(opt)
Bias / DC operating point (Fig. 14)	SMU / DC source-meter	I–V sweeps; VCE, ICE, supply-current, and bias-point verification

Table 5. Correspondence between the simulated analyses and the bench instruments that measure the same quantities.

12.2 Calibration, De-embedding and extraction

Accurate RF measurement begins by removing systematic setup errors. A VNA measurement is preceded by SOLT (short-open-load-thru) or TRL (thru-reflect-line) calibration to move the reference plane to the device terminals. Fixtured or on-wafer measurements also require de-embedding of pad, probe, fixture, and interconnect parasitics using open/short or thru structures. The measured reflection coefficient relates to impedance by Γ, and Gain, return loss, and stability are then extracted from calibrated S-parameters.

Γ is calculated as

$$\Gamma = (ZL - Z0)/(ZL + Z0) \qquad (14)$$

where $Z_0$is typically 50 Ω. Gain, return loss, matching quality, and stability are then extracted from the calibrated S-parameters.

12.3 Linearity and noise measurement

Linearity is characterized with two calibrated signal generators combined into the device and a spectrum analyzer measuring the IM3 products; the intercept follows from Eq. (5), matching the ADS two-tone simulation method. Noise figure can be measured with the Y-factor hot/cold-source method using a noise-figure analyzer, while NFmin, Rn, and Γopt require a noiseparameter or source-pull measurement setup. EVM and ACPR are measured using a vector signal analyzer and spectrum analyzer, respectively, closing the loop between the simulated system metrics and a measured prototype.

13. Consolidated Results and Conclusion

Table 6 consolidates the measured outcomes against the design objectives. The receiver chain meets a ≈ 76.1 dB gain at ≈ 4.2 dB cascaded noise figure; the amplifier is biased near V_{CE} ≈ 8.05 V / I_{CE} ≈ 2.03 mA, model-verified against reference S-parameters, stabilized to K > 1, noise-matched to NF_{min} ≈ 1.74 dB near 1.86 GHz, and impedance-matched with the synthesized L-C network.

Metric	**Result**	**Basis**
Adjacent-channel leakage	≈ 50–60 dBc	ACPR spectrum, Eq. (1)
EVM (peak)	≈ 0.011 norm.	Constellation, Eq. (2)
Receiver gain	≈ 76.1 dB	Cascaded budget
Receiver NF	≈ 4.2 dB (3.7 dB by Friis)	Friis, Eq. (3); Table 3
Two-tone linearity	OIP3 ≈ 21.8 dBm, IIP3 ≈ −25.7 dBm	Eq. (5–6); 70/72 MHz
Bias point	VCE ≈ 8.05 V, ICE ≈ 2.03 mA	Optimized bias network
Minimum noise figure	≈ 1.74 dB @ ~1.86 GHz	Noise analysis, Eq. (12)
Stability	K > 1 (stabilized)	Rollett, Eq. (10–11)
Matching network	Cin/Lin/Lout/Cout + stab.	L-section synthesis, Eq. (13)

Table 6. Consolidated measured results against the design objectives.

The study traverses the complete RF design arc- from system budgets and modulated-signal linearity, through device biasing and model verification, to a noise-optimized, stabilized, impedance-matched amplifier- with the key results interpreted against closed-form RF relations and mapped to their bench-measurement counterparts. It demonstrates the core competency set for RF / microwave and PHY-layer hardware design: system-level linearity and budgeting, S-parameter and noise design, stability analysis, and impedance matching in Keysight ADS.

14. References

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